Monash University Malaysia
STEM Grant Scheme 2026 - Application Form

A. APPLICANTS DETAILS

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Co-Investigators *(Please attach CVs)*	**1. Dr. Alina Arulsamy (School of Medicine and Health Sciences, Monash University Malaysia)** **2. Dr. Sanoop Mallissery (School of Information Technology, Monash University Malaysia)** **3. Dr. Leong Shu Min (School of Information Technology, Monash University Malaysia)** **4. Assoc. Prof. Dr. Khoo Ching Soong (School of Medicine and Neurology, Universiti Kebangsaan Malaysia and Hospital Canselor Tuanku Muhriz UKM)**
Key Research Focus Areas	• **Artificial Intelligence and Data Science** • **Computational Modelling and Simulation**

A. RESEARCH APPLICATION DETAILS

Research project title

Automated Video Analysis of Facial and Body Expressions in Temporal Lobe Epileptic Seizures Using Multimodal Deep Learning

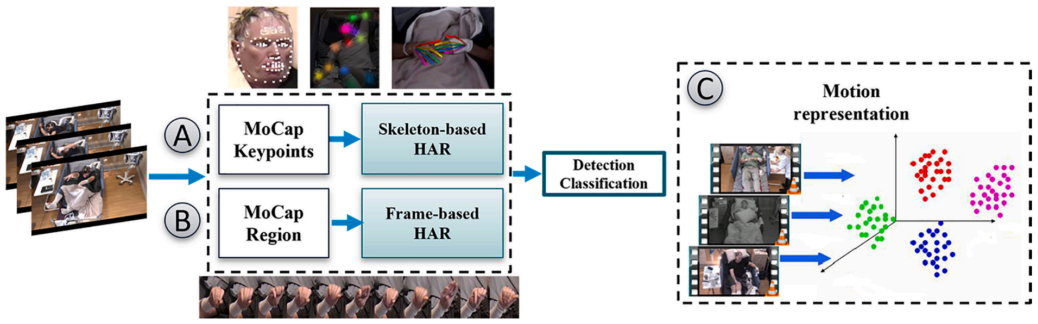


Figure 1: Overview of deep learning applications in seizure semiology analysis. (A) Motion capture and skeleton-based human action recognition. (B) Frame-based action recognition from vEEG recordings. (C) Feature representations of patient motion during seizures, facilitating similarity mapping and classification of epilepsy types.

Adapted from "Deep Learning Approaches for Seizure Video Analysis: A Review."

Executive summary (200 words max.)

Current clinical assessment of epileptic seizures in temporal lobe epilepsy (TLE) relies primarily on neurologists' visual interpretation of video-electroencephalogram (vEEG) recordings, which is subjective and prone to inter-rater variability. Subtle facial, emotional, and motor expressions that occur before and during seizures are often overlooked, limiting diagnostic precision and early intervention.

This project aims to develop an automated AI-driven vEEG analysis system that applies multimodal deep learning to detect, interpret, and predict these early behavioral patterns in TLE. Specifically, the objectives are to (1) design a computer vision framework capable of recognizing and correlating dynamic facial, affective, and motor cues in pre-ictal and ictal phases, and (2) validate the framework using ethically approved, clinician-annotated vEEG datasets from Hospital Canselor Tuanku Muhriz UKM (HUKM).

The proposed methodology integrates facial action coding, pose estimation, and temporal feature fusion using convolutional and transformer-based architectures, ensuring both explainability and clinical interpretability.

Expected outcomes include a validated AI-vEEG prototype, a Q1 publication or equivalent conference paper, and an ITEX submission to support external funding proposals. The research advances AI in healthcare and computational neurodiagnostics by enabling scalable, data-driven, and automated early seizure profiling, supporting Malaysia's precision medicine agenda and informing evidence-based policies to improve epilepsy care.

(200 words)

Problem statement

Epilepsy affects over 50 million individuals globally, with temporal lobe epilepsy (TLE) being one of the most prevalent and diagnostically challenging subtypes. Its heterogeneous semiology, often mimicking psychiatric or movement disorders, leads to frequent misdiagnosis, delayed treatment, and substantial socioeconomic burden. Despite advances in neuroimaging and electrophysiological monitoring, the behavioral characterization of seizures in TLE remains predominantly qualitative, relying on subjective interpretation rather than objective, data-driven metrics.

Traditional clinical assessment, particularly through video electroencephalogram (vEEG) recordings, depends heavily on manual visual review by neurologists. This introduces inter-rater variability, diagnostic inconsistency, and limits the ability to capture subtle, transient cues such as micro-expressions, facial emotion changes, and dystonic motor posturing that occur in pre-ictal and early ictal phases. These minute behavioral changes are analogous to early physiological alterations like heart rate variability before a seizure, are often present in vEEG data but remain undetectable without computational assistance.

Currently, there is a critical gap in scalable, automated analytic frameworks capable of integrating multimodal behavioral signals, such as facial action units, body pose dynamics, and motion trajectories, into quantifiable clinical markers. The absence of such systems hampers objective evaluation of disease progression, treatment efficacy, and rehabilitation outcomes, thereby constraining both clinical decision-making and research advancement.

Addressing this gap through artificial intelligence (AI)-assisted vEEG analysis offers the potential to transform clinical workflows. By leveraging multimodal deep learning, this project aims to develop a framework that automatically detects and interprets behavioral signatures associated with TLE. Such a system could enable standardized, near–real-time quantification of facial–emotion–motor interactions, enhance diagnostic accuracy, support early seizure prediction, and streamline longitudinal monitoring.

Ultimately, this research seeks to advance the field of computational neurodiagnostics by establishing objective, reproducible, and explainable AI-driven insights into seizure manifestations, bridging the existing divide between neurological observation and precision medicine in epilepsy care.

Research objectives

1. **To design and develop an automated multimodal AI framework** that integrates computer vision and deep learning techniques to recognize, classify, and correlate dynamic facial, emotional, and motor cues captured in pre-ictal and ictal phases of temporal lobe epilepsy (TLE) from video-electroencephalogram (vEEG) recordings.

2. **To validate the developed framework** using **clinician-annotated vEEG datasets** from Hospital Canselor Tuanku Muhriz UKM (HUKM), employing patient-independent data splits to ensure generalizability and clinical robustness.

3. **To establish objective, video-derived biomarkers for TLE seizure prediction and detection** by quantifying model performance through metrics such as AUROC, F1-score, false positives per hour, and onset/offset localization accuracy, thereby enabling clinical interpretability and integration into routine neurodiagnostic workflows.

Literature review

Temporal lobe epilepsy (TLE) represents one of the most common focal epilepsy types globally, often accompanied by significant but heterogenous emotional and motor manifestations due to its involvement of limbic and temporal networks. Emotional dysregulation, altered facial expressions, and dystonic (motor) posturing are frequent manifestations during seizures, reflecting abnormal connectivity between temporal, limbic, and motor cortices. Yet, these complex behavioural phenomena are typically assessed through subjective clinical observation, leading to variability in interpretation and underestimation of their diagnostic potential. Blair (2012) suggested that seizure auras and automatism are specific features in TLE and that a careful understanding and interpretation of this unique seizure semiology can help diagnose, and possibly predict seizure onset. Similarly and more recently, Yang *et al.* (2022) showcased that there are distinct semiologic patterns of automatisms between different epilepsy types which, if automated, can help neurologists diagnose more accurately and efficiently. Unfortunately, similar research on facial and emotional expressions in TLE or other epilepsy types are scarce but are still worth investigating. This highlights the importance of facial, emotion and motor manifestation analysis as potential biomarkers of epileptogenic network activity.

Conventional seizure monitoring through clinical vEEG remains the gold standard but is limited by manual review, inter-rater variability, and lack of standardized quantification. Behavioural signs such as facial twitching, automatisms, or affective displays are often noted descriptively rather than objectively quantified. Moreover, minute changes of these behavioural signs before a seizure episode may not be visible to the naked eye, and thus go unnoticed. To address these limitations, computer vision-based approaches have emerged as powerful tools for automatic behavioural characterization and prediction. Ahmedt-Aristizabal *et al.* (2019) demonstrated the feasibility of using vision-based methods to analyse seizure semiology, identifying characteristic movement and postural changes associated with different seizure types. Similarly, Hou *et al.* (2022) applied automated video analysis to capture emotional and dystonic expressions during epileptic seizures, illustrating that AI can detect subtle motor and affective features beyond human perception and warrants further research. However, these early frameworks largely relied on unimodal data and non-specific seizures (not TLE), limiting their generalizability and interpretability across heterogeneous clinical datasets. Nevertheless, these early works in automated video analysis justifies the importance and current relevance of research into AI automation for seizure detection and prediction using facial, emotion and motor cues.

Recent developments in multimodal deep learning have shown promise in decoding complex human behaviours through the integration of visual, auditory, and physiological cues. Baltrušaitis *et al.* (2018) provided a comprehensive taxonomy for multimodal machine learning, outlining methods for fusing diverse data streams to improve robustness and contextual understanding. Extending these principles, Chaabene *et al.* (2025) emphasized the potential of multimodal fusion in healthcare, noting that combining spatial, temporal, and affective features enhances diagnostic accuracy. In the context of epilepsy, integrating facial expression dynamics, emotion cues, and motor movements with EEG data could yield powerful biomarkers for predicting seizure onset and classifying seizure types, particularly in TLE, where emotional and behavioural alterations precede electrophysiological changes.

To ensure clinical utility, explainability remains a cornerstone of medical AI development. Ching *et al.* (2018) and Shuaib (2024) underscored that model transparency and interpretability are critical for clinician trust, ethical deployment, and regulatory compliance. AI frameworks enable visualization of decision-making processes and identification of salient features, which are essential for validating automated predictions against expert neurophysiological interpretation. Incorporating these elements into an AI framework for epilepsy ensures that algorithmic outputs can be correlated with clinically meaningful cues, such as preictal (before seizure onset) facial tension or emotion shifts observable in vEEG footage.

Despite these advances, a significant translational gap persists between neuroscience insights into emotional-motor dysfunction in TLE and automated AI-based assessment and prediction methods. Current literature lacks validated multimodal frameworks capable of recognizing and correlating dynamic emotion and motor cues with seizure activity across large, clinically annotated vEEG datasets. The present study aims to address these gaps by designing, developing and validating an automated multimodal AI framework capable of recognizing, classifying, and correlating dynamic facial, emotional, and motor cues before and during seizures in TLE. This will allow quantifiable behavioural biomarkers to be identified which will enhance seizure prediction and detection accuracy, bridging the gap between neurobehavioral science and computational diagnostics, and ultimately advancing precision monitoring and patient care in TLE, and later to other types of epilepsies, across different ages.

Reference List

Ahmedt-Aristizabal, D., Denman, S., Nguyen, K., Sridharan, S., Dionisio, S., & Fookes, C. (2019). Understanding patients' behavior: Vision-based analysis of seizure disorders. IEEE Journal of Biomedical and Health Informatics, 23(6), 2583–2591. DOI: 10.1109/JBHI.2019.2895855

Ahmedt-Aristizabal, D., Armin, M. A., Hayder, Z., Garcia-Cairasco, N., Petersson, L., Fookes, C., ... & McGonigal, A. (2024). Deep learning approaches for seizure video analysis: A review. Epilepsy & Behavior, 154, 109735. DOI: https://doi.org/10.1016/j.yebeh.2024.109735

Baltrušaitis, T., Ahuja, C., & Morency, L. P. (2018). Multimodal machine learning: A survey and taxonomy. IEEE Transactions on Pattern Analysis and Machine Intelligence, 41(2), 423–443.
Blair R. D. (2012). Temporal lobe epilepsy semiology. Epilepsy research and treatment, 2012, 751510. DOI: 10.1109/TPAMI.2018.2798607

Chaabene, S., Boudaya, A., Bouaziz, B., & Chaari, L. (2025). An overview of methods and techniques in multimodal data fusion with application to healthcare. International Journal of Data Science and Analytics, 1–25. DOI: 10.1007/s41060-025-00715-0

Ching, T., Himmelstein, D. S., Beaulieu-Jones, B. K., Kalinin, A. A., Do, B. T., Way, G. P., & Greene, C. S. (2018). Opportunities and obstacles for deep learning in biology and medicine. Journal of the Royal Society Interface, 15(141), 20170387. DOI: 10.1098/rsif.2017.0387

Hou, J. C., Thonnat, M., Bartolomei, F., & McGonigal, A. (2022). Automated video analysis of emotion and dystonia in epileptic seizures. Epilepsy Research, 184, 106953. DOI: 10.1016/j.eplepsyres.2022.106953

Shuaib, A. (2024). Transforming healthcare with AI: Promises, pitfalls, and pathways forward. International Journal of General Medicine, 17, 1765–1771. DOI: 10.2147/IJGM.S449598

Yang, B., Mo, J., Zhang, C., Wang, X., Sang, L., Zheng, Z., Gao, D., Zhao, X., Wang, Y., Liu, C., Zhao, B., Guo, Z., Shao, X., Zhang, J., Zhang, K., & Hu, W. (2022). Clinical features of automatisms and correlation with the seizure onset zones: A cluster analysis of 74 surgically-treated cases. Seizure, 94, 82–89. DOI: 10.1016/j.seizure.2021.11.015

Research methodology

The proposed study will be implemented over 12 months (January–December 2026) through four structured phases, encompassing data curation, AI model development, validation, and translation. The methodology ensures scientific rigor, ethical compliance, and clinical applicability.

Phase 1 – Data Curation and Preprocessing

Objectives:
To curate, anonymize, and preprocess multimodal clinical vEEG recordings for AI model development.

Activities:

- **Data acquisition**: Obtain ethics-approved vEEG recordings from Hospital Canselor Tuanku Muhriz (HUKM) under existing neurology research collaboration.
- **Data Anonymisation**: Apply automated face masking and metadata de-identification protocols to ensure patient confidentiality for publications.
- **Feature extraction**:
    - Facial features: Action Unit (AU) extraction, micro-expression detection, and facial landmark tracking.
    - Body motion features: 2D skeletal keypoint estimation per frame with optional 3D temporal lifting and kinematic trajectory analysis.
    - Motion dynamics: Optical flow and temporal gradient computation for movement quantification.
- **Dataset preprocessing and partitioning**: Implement patient-independent train/validation/test splits to prevent overfitting. Multi-camera or ward-based splits will be used where applicable.
- **Notable auxiliary datasets**: Public EEG or multimodal datasets (e.g., University of California Irvine Epileptic Seizure Recognition Dataset[1], Children's Hospital Boston-MIT Dataset[2], Bangalore EEG Epilepsy Dataset[3]) will serve as supplementary training resources for model generalization and pretraining.

Deliverable: Curated, preprocessed, and annotated dataset compliant with Monash ethics and clinical governance standards.

Phase 2 – Multimodal AI Framework Development

Objectives:
To develop a multimodal deep-learning architecture that integrates facial, motor, and temporal features for seizure detection and prediction.

Activities:

- **Model architecture**:
    - Branch-specific encoders (e.g., CNN for spatial features, Vision Transformer for temporal fusion).
    - Comparative experiments between early fusion (concatenated feature tokens) and late fusion (cross-modal attention-based integration).
- **Temporal modeling**:
    - Implement temporal convolutional networks (TCN) or transformer decoders for event boundary localization and onset/offset detection.
- **Pretraining**: Leverage emotion and pose datasets (e.g., BP4D, CASME II, SAMM, NTU RGB+D) for transfer learning.
- **Explainability**: Integrate Grad-CAM and attention heatmaps for interpretable AI outputs,

[1] https://archive.ics.uci.edu/datasets
[2] https://physionet.org/content/chbmit/1.0.0/
[3] https://archive.ics.uci.edu/dataset/1134/

allowing clinicians to visualize decision rationale.

Deliverable: Prototype multimodal AI model capable of identifying pre-ictal and ictal behavioral cues in vEEG recordings.

Phase 3 – Model Validation and Correlation Analysis

Objectives:
To evaluate the model's clinical validity and establish quantifiable behavioral biomarkers.

Activities:

- **Performance evaluation**:
  - Compute sensitivity, specificity, AUROC, F1-score, false positives per hour, and onset/offset localization error.
  - Assess model generalizability through patient-independent validation.
- **Agreement analysis**:
  - Compare AI outputs with clinician annotations and compute Cohen's κ for inter-rater reliability.
- **Correlation studies**:
  - Quantify associations between AI-derived emotion/motor scores and clinical severity indicators (e.g., seizure lateralization, medication response).

Deliverable: Validated AI-vEEG seizure detection and prediction framework with clinically interpretable performance metrics.

Phase 4 – Translation, Dissemination, and Knowledge Transfer

Objectives:
To translate research outcomes into tangible clinical and academic outputs, supporting Monash's strategic goals in AI and Health.

Activities:

- Develop a **clinician-oriented prototype interface** with explainable overlays to facilitate real-time seizure interpretation.
- Conduct a **usability evaluation** with neurologists (measuring time efficiency, trust, and interpretability).
- Prepare and submit at least one Q1 journal article, one conference paper, and an ITEX/MTE invention disclosure.
- Formulate an external grant proposal (e.g., FRGS 2027 or MOHE Translational Grant and/or Wellcome Trust) for large-scale validation.

Deliverables:
Validated AI-vEEG prototype, scholarly publications, intellectual property disclosure, and readiness for external funding application.

Research Flowchart

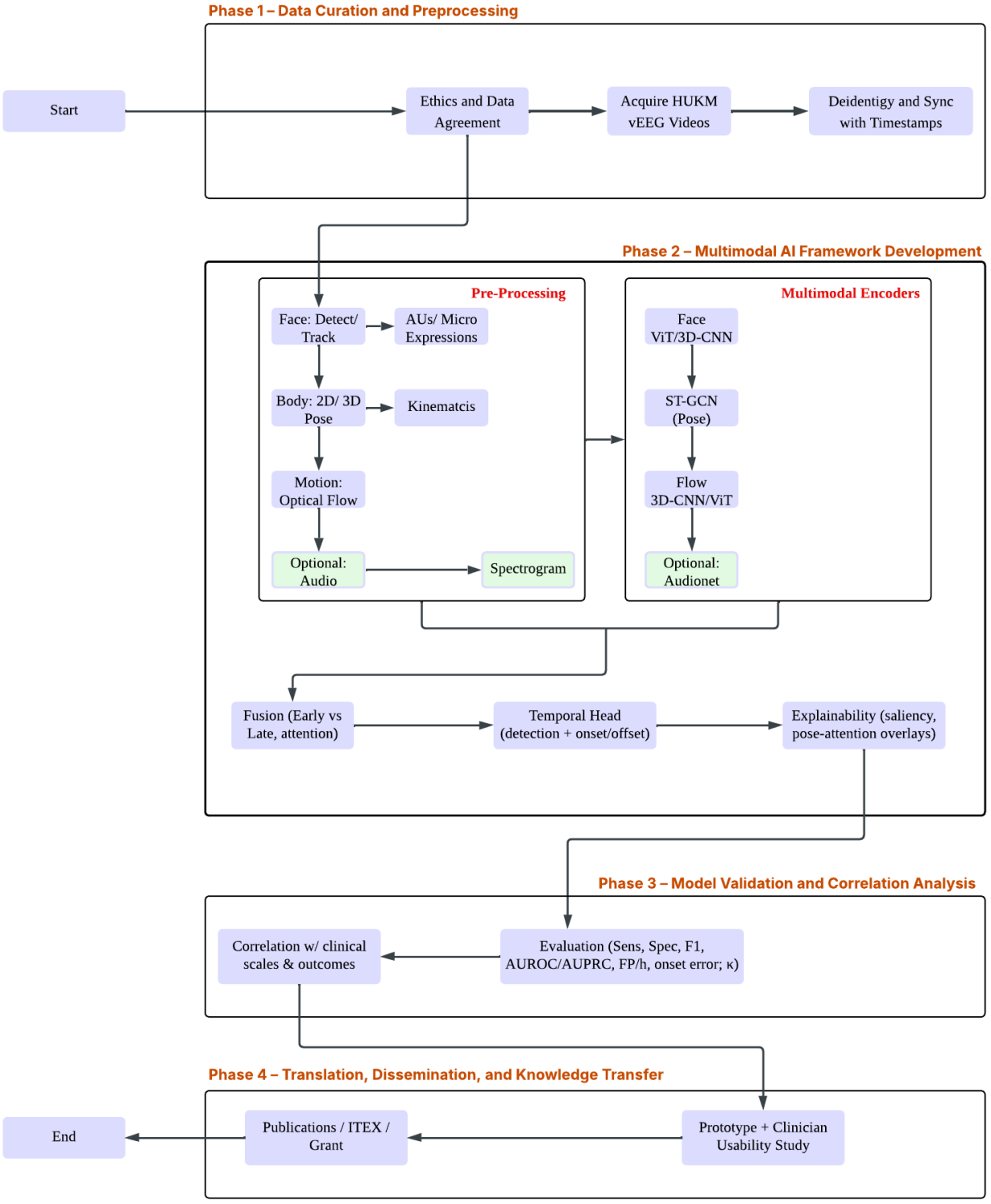


Figure 2: Research Flowchart

Contribution to knowledge and originality (400 words max.)

This research addresses a critical gap in epilepsy diagnosis and treatment monitoring by introducing an innovative, AI-driven approach that combines affective computing with precision neurological assessment and prediction. While previous studies have relied heavily on subjective seizure diaries, caregiver observations, or traditional EEG/vEEG interpretations by neurologists, these methods often lack objectivity, consistency, and scalability. Existing automated analyses typically focus on either EEG signal patterns or isolated behavioral metrics, without incorporating comprehensive multimodal video data. Consequently, subtle but clinically significant emotional and motor features associated with temporal lobe epilepsy often go undetected or undervalued in today's clinical setting. Our project pioneers a novel integration of vision transformers and affective computing to perform dynamic, automated video EEG analysis. This allows for quantitative assessment of facial expressions, emotional states, and dystonic motor signs that occur during seizures, elements critical for accurate classification and understanding of epilepsy types.

This work fills a significant knowledge gap in epilepsy diagnostics: the lack of an automated system capable of integrating facial, emotional and motor data with clinical EEG signals. It supports neurologists in delivering more precise, personalized assessments, which can improve treatment decisions and management as well as patient outcomes. In addition, tools to predict seizures have been a recent focus in the epilepsy field, where studies have looked at changes in Heart Rate Variability (HRV) as a predictive biomarker before seizure onset. However, the limitations of that approach include significant variability and non-stationarity in HRV signals, meaning that the patterns preceding seizures can vary greatly both between individuals and between seizures in the same individual, possibly resulting in high false-positives. Thus, by using AI-vEEG to recognize multi-biomarkers such as face, emotion, and motor expressions and make predictions, the system can provide a more comprehensive and nuanced understanding of seizure dynamics beyond EEG signals alone. This multimodal approach enables early and precise seizure detection by capturing subtle behavioral and emotional changes that often precede or accompany epileptic events. Furthermore, integrating these biomarkers with AI enhances the model's predictive accuracy and robustness, potentially reducing false alarms and enabling timely intervention, thereby improving patient safety and quality of life.

Academically, the research contributes novel methodologies combining AI and clinical neuroscience, fostering interdisciplinary advancement. Clinically, it promises to transform epilepsy care by establishing a data-driven diagnostic paradigm that is both accessible and effective, empowering future healthcare providers and benefiting communities living with epilepsy.

(394 words)

Strategy alignment and complementarity (500 words max.)

This project demonstrates clear alignment with the **United Nations Sustainable Development Goals (SDGs), particularly SDG 3 – Good Health and Well-being and SDG 9 – Industry, Innovation, and Infrastructure**, while directly supporting Monash University Malaysia's strategic priorities in Artificial Intelligence (AI) and Digital Health.

Under SDG 3 (Target 3.4), which focuses on reducing premature mortality from non-communicable diseases through prevention, treatment, and mental health promotion, this project addresses a pressing healthcare challenge - the accurate and timely diagnosis of epilepsy, particularly Temporal Lobe Epilepsy (TLE). Epilepsy is a chronic neurological disorder that profoundly affects quality of life and, without proper diagnosis and management, contributes significantly to global disease burden. Current assessment methods are predominantly manual and qualitative, relying on neurologists' visual interpretation of video electroencephalogram (vEEG) recordings. This introduces inter-rater variability, diagnostic inconsistency, and delayed intervention. By developing an AI-driven vEEG analysis framework, this project enhances clinical objectivity, precision, and accessibility in epilepsy care. The proposed system employs multimodal deep learning to automatically recognize, interpret, and predict seizure-related facial, emotional, and motor cues, providing neurologists with real-time, evidence-based insights. This innovation has the potential to improve treatment planning, facilitate early seizure prediction, and enhance patient safety and mental well-being. Through these outcomes, the project contributes directly to improving neurological health equity in Malaysia and beyond.

Aligned with SDG 9, the project promotes sustainable healthcare innovation by strengthening the integration of AI and data science within national healthcare infrastructure. The AI-vEEG framework introduces a scalable, data-driven diagnostic model that can be integrated with hospital information systems, remote monitoring solutions, and future wearable technologies. This fosters the development of resilient, technology-enabled healthcare systems and supports Malaysia's National AI Roadmap (2021–2027) and Digital Health Transformation Strategy. Moreover, the project's emphasis on prototype development and translational readiness positions it as a catalyst for industry collaboration, intellectual property generation, and future commercialization through Monash's innovation ecosystem and potential partnerships with digital health startups.

The research team composition exemplifies strong interdisciplinarity and complementarity:

- Dr. Yin Yin Low (School of Information Technology, MUM) leads the project's technical direction, contributing expertise in AI, multimodal machine learning, and affective computing. She will oversee framework design, model development, and technical validation.
- Dr. Alina Arulsamy (School of Medicine and Health Sciences, MUM) provides clinical and neuroscientific expertise, with specialized research in epilepsy and access to HUKM's vEEG dataset for model training and validation.
- Assoc. Prof. Dr. Khoo Ching Soong (Universiti Kebangsaan Malaysia / HUKM) ensures clinical oversight, ethical governance, and translational relevance through direct neurological practice.
- Dr. Sanoop Mallissery and Dr. Leong Shu Min (School of Information Technology, MUM) contribute expertise in AI, multimodal machine learning, computational modeling, affective computing, security and system integration, ensuring methodological robustness and scalability.

Together, this interdisciplinary consortium bridges algorithmic innovation with clinical application, uniting technology and medicine to deliver real-world healthcare solutions. The project reinforces Monash University Malaysia's commitment to impactful, sustainable, and translational research, contributing to Malaysia's national health goals while advancing global leadership in AI-driven precision medicine.

(485 words)

Impact statement

Academia	This research strengthens Monash University Malaysia's leadership at the intersection of **artificial intelligence, affective computing, and clinical neuroscience**. It will advance the university's strategic agenda in AI-driven healthcare innovation through the development of a validated, explainable multimodal framework for neurological diagnostics. The project will yield **high-impact Q1 journal and conference publications**, alongside **ITEX and innovation showcase submissions**, enhancing the campus's research visibility and reputation. Furthermore, it establishes a foundation for external grant applications such as FRGS and MOHE Translational Research Grants, contributing to research income growth. The interdisciplinary collaboration between the Schools of Information Technology, Jeffrey Cheah School of Medicine and Health Sciences and Universiti Kebangsaan Malaysia will also support research capacity building, postgraduate training, and long-term academic partnerships with Malaysian hospitals and international institutions.
Society	For patients and caregivers, this project directly enhances **quality of life, safety, and independence**. The AI-vEEG framework enables early and objective detection of emotional and motor changes associated with temporal lobe epilepsy, supporting more precise treatment, rehabilitation, and counseling strategies. The system's predictive capability allows clinicians to anticipate seizures, enabling preemptive measures that reduce injury risk and anxiety. In the long term, this innovation could facilitate greater social inclusion for individuals with epilepsy, including evidence-based assessments for daily living activities such as driving, employment, and self-management.
Industry	The project introduces a **novel AI-based clinical video analytics framework** with strong commercialization potential in the **healthtech and telemedicine sectors**. By translating research outputs into prototype solutions, the study supports the development of intelligent neurological monitoring systems for hospitals, neurocentres, and digital health startups. The AI-vEEG prototype can be extended to other disorders involving behavioral or motor dysregulation, broadening its market application. Through Monash's innovation ecosystem, the project fosters technology transfer, intellectual property creation, and future industry-academic collaborations that advance Malaysia's digital health economy.
Government	This research aligns with national initiatives such as the **National Artificial Intelligence Roadmap**, **Malaysia's Digital Health Transformation Strategy, and National Health Digitalisation Framework (One Patient, One Health Record)**. The project contributes to the government's vision of precision healthcare through AI-enabled early diagnosis and continuous monitoring. In the longer term, integration of the AI-vEEG model into national healthcare and smart mobility frameworks could inform transportation policy reform for individuals with controlled epilepsy, promoting safe, equitable participation in daily activities while supporting public safety objectives.
Environment	By leveraging automated, data-driven video analytics, this research reduces dependence on repeated manual assessments and physical monitoring sessions. Digital clinical workflows minimize travel, hospital visits, and redundant testing, resulting in a lower carbon footprint. Moreover, the project's use of cloud-based computing resources supports environmentally sustainable, paperless research operations consistent with Monash University's Net Zero Initiative and commitment to sustainable digital innovation.

Milestones

No.	Milestones	Key deliverables / Project activities	Date to be completed
Phase 1 – Data Curation and Preprocessing			
1.	Project Setup, Ethics Approval, Data Curation & Pilot Annotations	● Obtain ethics clearance from HUKM and MUMHREC and finalize Data Use Agreement (DUA) with HUKM. ● Establish data governance SOPs for secure handling of vEEG data. ● Develop and deploy an automated (or semi-automated) de-identification pipeline to perform face masking, metadata stripping, and contextual anonymization for patient confidentiality. ● Conduct initial data preprocessing and annotation to produce a curated, ethically compliant vEEG dataset (v0.1) suitable for AI training.	31 March 2026 (Month 3)
Phase 2 – Multimodal AI Framework Development			
2.	Facial–Emotion Model Development (Stage 1)	● Implement facial Action Units (AU) and micro-expression recognition modules. ● Conduct pretraining, ablation studies, and clinician feedback round #1. ● Evaluate performance using sensitivity, F1-score, and AUROC metrics.	30 May 2026 (Month 5)
3.	Motor (Pose) Model Development & Temporal Boundary Detection (Stage 2)	● Develop 2D pose estimation and temporal boundary detection networks. ● Conduct clinician feedback round #2 and refine feature correlations.	31 August 2026 (Month 8)
Phase 3 – Model Validation and Correlation Analysis			
4.	Multimodal Integration & Validation (Stage 3)	● Integrate facial and motor modalities using early/late fusion with transformer-based fusion. ● Conduct patient-independent validation and reader study with clinicians. ● Generate validation reports and clinical correlation results.	31 October 2026 (Month 10)

Phase 4 – Translation, Dissemination, and Knowledge Transfer			
5.	Prototype Development, Usability Evaluation & Project Close-Out	● Deliver AI-vEEG Prototype v1.0 with clinician explainability overlays. ● Conduct usability pilot with neurologists at HUKM (assessing efficiency, interpretability, and trust). ● Finalize project documentation, model card, and final report.	31 December 2026 (Month 12)

Gantt Chart (Jan – Dec 2026)

Milestones / Activities	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Phase 1 – Data Curation and Preprocessing												
Project setup, ethics approval & collaboration finalization												
Data governance & de-identification pipeline												
Phase 2 – Multimodal AI Framework Development												
Facial–Emotion Model Development (Stage 1)												
Motor (Pose) Model Development & Temporal Boundary Detection (Stage 2)												
Phase 3 – Model Validation and Correlation Analysis												
Multimodal Integration & Validation (Stage 3)												
Phase 4 – Translation, Dissemination, and Knowledge Transfer												
Prototype Development, Usability Evaluation & Project Close-Out												
Publication drafting, dissemination & reporting												

Research output

a) Publications:
The project will produce a minimum of one Q1 journal publication in a high-impact outlet such as Epilepsia, NeuroImage, or IEEE Transactions on Affective Computing. Additional outputs will include one international conference paper in AI or healthcare informatics (e.g., ICIP, EMBC, or MICCAI) to disseminate findings to the broader scientific community.

b) External research grant for further research:
Project outcomes will serve as the foundation for subsequent external funding applications, targeting:

- Fundamental Research Grant Scheme (FRGS) 2027 – Ministry of Higher Education (MOHE)
- MOHE Translational Research Grant (2027)
- International opportunities such as the Bill & Melinda Gates Foundation and the Wellcome Trust (Digital Health and AI for Healthcare Programs)

c) Intellectual Property and Commercialisation:
The project will deliver a prototype AI-based clinical video analytics system (AI-vEEG) with potential for intellectual property registration under Monash University Malaysia's Innovation and Commercialisation framework. The prototype will demonstrate clinical relevance, scalability, and readiness for commercialization through strategic industry partnerships and licensing opportunities.

d) Industry Collaboration:
This research will foster long-term collaboration with healthcare stakeholders, including clinical institutions, neurology centres, and healthcare AI startups. The project team aims to initiate a partnership discussion with the Ministry of Health Malaysia (MOH) for potential integration of the AI-vEEG system into clinical workflows and national health digitalisation initiatives.

e) Others:
The innovation will be showcased through submissions to major invention and innovation platforms, including the International Invention, **Innovation and Technology Exhibition (ITEX)** and the **Malaysia Technology Expo (MTE)**. These platforms will further strengthen Monash University Malaysia's visibility in translational AI healthcare research and technology-driven societal impact.

Future plan (500 words max.)

Upon completion, the findings and prototype model (AI-vEEG) developed through this project will serve as a robust foundation for a large-scale, multi-site clinical validation study. The next research phase will involve collaboration with multiple hospitals across Malaysia, particularly neurology departments with established video-electroencephalogram (vEEG) facilities. This multi-institutional validation will enable the collection of a diverse and representative dataset encompassing various patient demographics, seizure types, and healthcare environments. Such an expansion is critical to ensuring the reliability, clinical accuracy, and scalability of the AI-vEEG system for real-world implementation.

Following validation in Temporal Lobe Epilepsy (TLE), the model will be expanded and retrained to recognize and analyze other forms of epilepsy, including focal, generalized, and genetic epilepsy syndromes. Special focus will be given to pediatric epilepsy, which presents unique diagnostic challenges due to developmental and behavioral variability. This extension will involve computational adaptation and dataset enhancement to accommodate different electrophysiological and behavioral characteristics, thereby improving the system's generalizability and clinical versatility across a wider patient spectrum.

In parallel, the project will explore integration with wearable and remote EEG monitoring devices to enable continuous, ambulatory seizure tracking and early warning systems for home and community settings. Future iterations may also interface with smart mobility or vehicle safety systems, allowing automated detection and mitigation measures (e.g., caregiver alerts or emergency responses) during seizure events, contributing to patient safety and independence.

A key translational objective is the integration of the AI-vEEG prototype into existing hospital information systems and electronic medical records (EMRs). This will facilitate seamless incorporation into clinical workflows, providing neurologists with real-time decision support tools for diagnosis, treatment planning, and post-therapy monitoring. The team will work closely with clinical partners and health technology vendors to ensure compliance with medical data standards, cybersecurity, and interoperability requirements.

To sustain and scale this research, the team will strategically pursue national and international funding opportunities. Future grant applications will target the Ministry of Higher Education (MOHE) through the Fundamental Research Grant Scheme (FRGS) and Translational Research Grant, as well as the Ministry of Science, Technology and Innovation (MOSTI) under the Malaysia AI Roadmap 2027. Regionally, the team will engage with the ASEAN AI Healthcare Consortium to position the AI-vEEG framework as a regional benchmark for intelligent neurological diagnostics. Globally, international collaborations and funding from organizations such as the Bill & Melinda Gates Foundation and the Wellcome Trust will be pursued to expand impact across low- and middle-income countries where access to epilepsy specialists is limited.

Ultimately, the project aims to establish Monash University Malaysia as a regional leader in AI-powered neurodiagnostics, driving cross-sector collaboration between academia, healthcare institutions, and industry. The sustained research trajectory will not only enhance Malaysia's capacity in digital health innovation but also contribute to a more equitable, data-driven, and patient-centric healthcare ecosystem.

(459 words)

Option 1: Application based
Output: Data Pipeline, Server Hosting, AI Application for analyzing Seizure

Option 2: Research based
Research Goal 1: Achieving accuracy fusion model, simple interface
Research Goal 2: Achieving adversarial robustness for fusion model, simple interface